



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

End-to-end data pipeline covering API data extraction, web scraping, exploratory SQL analytics, Interactive visualization, and supervised machine learning classification

- Summary of all results
- Landing success rate rose from near 0% in 2013 to over 80% by 2020
- Launch site and payload mass are both correlated with landing outcome
- The Decision Tree classifier was the best model

Introduction

- **Project background and context**
- SpaceX charges ~\$62M per launch vs. \$165M+ industry average, driven by Falcon 9 booster reusability.
- Predicting first-stage landing outcomes helps estimate actual launch costs and hardware savings.
- Public launch data and ML algorithms enable reliable binary prediction of landing success.
- **Problems you want to find answers**
- Which launch sites achieve the highest landing success rates?
- How do payload mass and orbit type impact successful booster recovery?
- Which ML classification model predicts landing outcomes with the highest accuracy?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
- Perform data wrangling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Performed predictive analysis covering standardization, train/test split, GridSearchCV across 4 classifiers

Data Collection

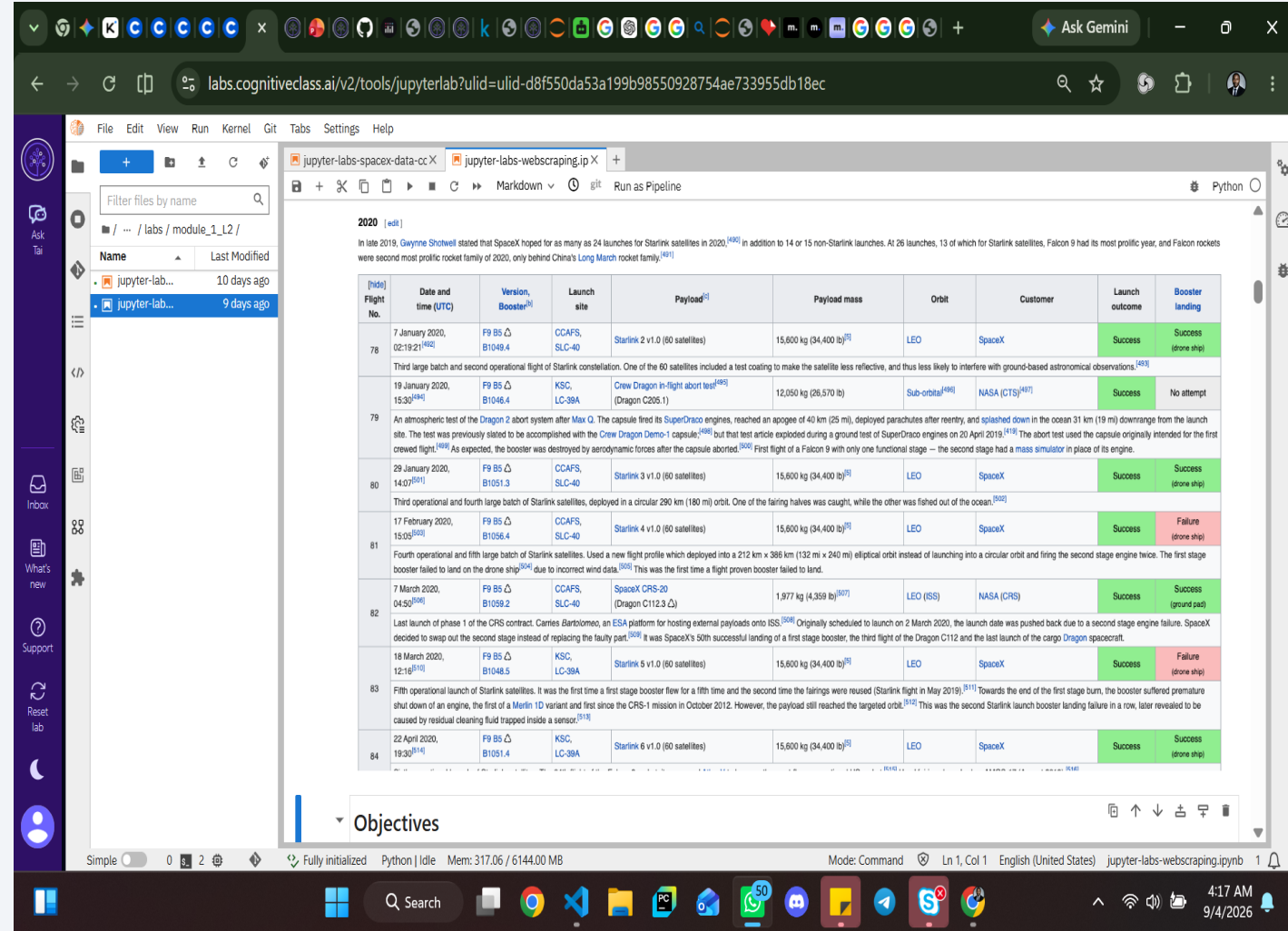
- SpaceX REST API: Extracted different features directly from SpaceX's official API endpoints.
- Wikipedia Web Scraping: Scraped landing outcomes and Falcon 9 launch logs using BeautifulSoup to capture additional historical launch context and features.

Data Collection - Scraping

Web scraped Falcon 9 launch records with BeautifulSoup:

- Extract a Falcon 9 launch records HTML table from Wikipedia
- Parse the table and convert it into a Pandas data frame

[Github link](#)



The screenshot shows a JupyterLab environment with a web browser displaying a table of Falcon 9 launch records for the year 2020. The table is titled "2020" and contains the following data:

Flight No.	Date and time (UTC)	Version, Booster ^[1]	Launch site	Payload ^[2]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21 ^[462]	F9 B5 Δ B1048.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[1]	LEO	SpaceX	Success	Success (drone ship)
79	19 January 2020, 15:30 ^[464]	F9 B5 Δ B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test ^[465] (Dragon C205.1)	12,050 kg (26,570 lb)	Sub-orbital ^[466]	NASA (CTS) ^[467]	Success	No attempt
80	29 January 2020, 14:07 ^[461]	F9 B5 Δ B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[1]	LEO	SpaceX	Success	Success (drone ship)
81	17 February 2020, 15:05 ^[468]	F9 B5 Δ B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[1]	LEO	SpaceX	Success	Failure (drone ship)
82	7 March 2020, 04:50 ^[469]	F9 B5 Δ B1059.2	CCAFS, SLC-40	SpaceX CRS-20 (Dragon C112.3 Δ)	1,977 kg (4,359 lb) ^[467]	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
83	18 March 2020, 12:16 ^[470]	F9 B5 Δ B1048.5	KSC, LC-39A	Starlink 5 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[1]	LEO	SpaceX	Success	Failure (drone ship)
84	22 April 2020, 19:30 ^[471]	F9 B5 Δ B1051.4	KSC, LC-39A	Starlink 6 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[1]	LEO	SpaceX	Success	Success (drone ship)

The interface also shows a file explorer on the left with files named "jupyter-lab..." and a terminal at the bottom with the text "Objectives".

Data Wrangling

This phase had two main objectives;

- Exploratory Data Analysis - to find some patterns in the data
- Determining Training Labels- to determine what would be the label for training supervised models.

GitHub URL

EDA with Data Visualization

In order to achieve an Interactive Visual Analytics for geo-launch sites, I used folium to achieve;

- 1: Mark all launch sites on a map
- 2: Mark the success/failed launches for each site on the map
- 3: Calculate the distances between a launch site to its proximities

Plotly Dash was also used to create an interactive visual dashboard with a pie chart and a scatter point chart.

[Github URL](#)

EDA with SQL

I performed SQL queries that;

- Display the total payload mass carried by boosters launched by NASA (CRS)
- List the date when the first successful landing outcome in ground pad was achieve
- List the total number of successful and failure mission outcomes
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[Github URL](#)

Build an Interactive Map with Folium

Summary

- Added a Circle + text Marker for each of the 4 launch sites, and grouped individual launch records with MarkerCluster, color-coded green (success) / red (failure)
- Added MousePosition to read exact coordinates while exploring the map, used to identify nearby coastlines, cities, railways and highways
- Calculated great-circle distances (Haversine formula) from a launch site to these points of interest and visualized them with PolyLine connectors

Predictive Analysis (Classification)

- Trained and tuned four models with GridSearchCV and a cross-validation of 10
Models - Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors
- Standardized my features with StandardScaler

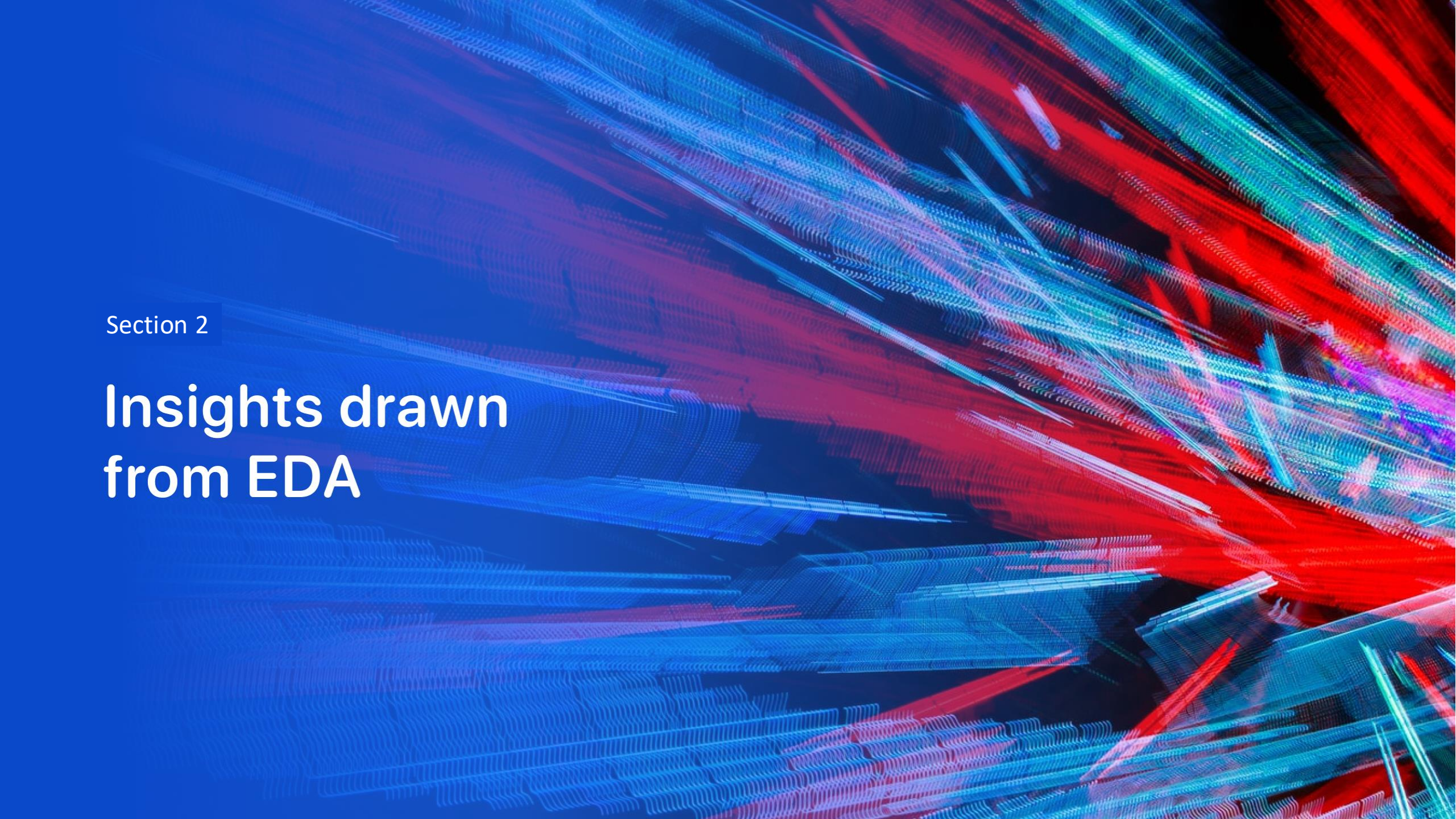
[Github URL](#)

Results

- Predictive analysis results

The Decision Tree classifier model is the most reliable of the models tested with 90.1% accuracy.

KSC LC-39A shows both a large number of successful launches and the highest success ratio among all four launch sites.

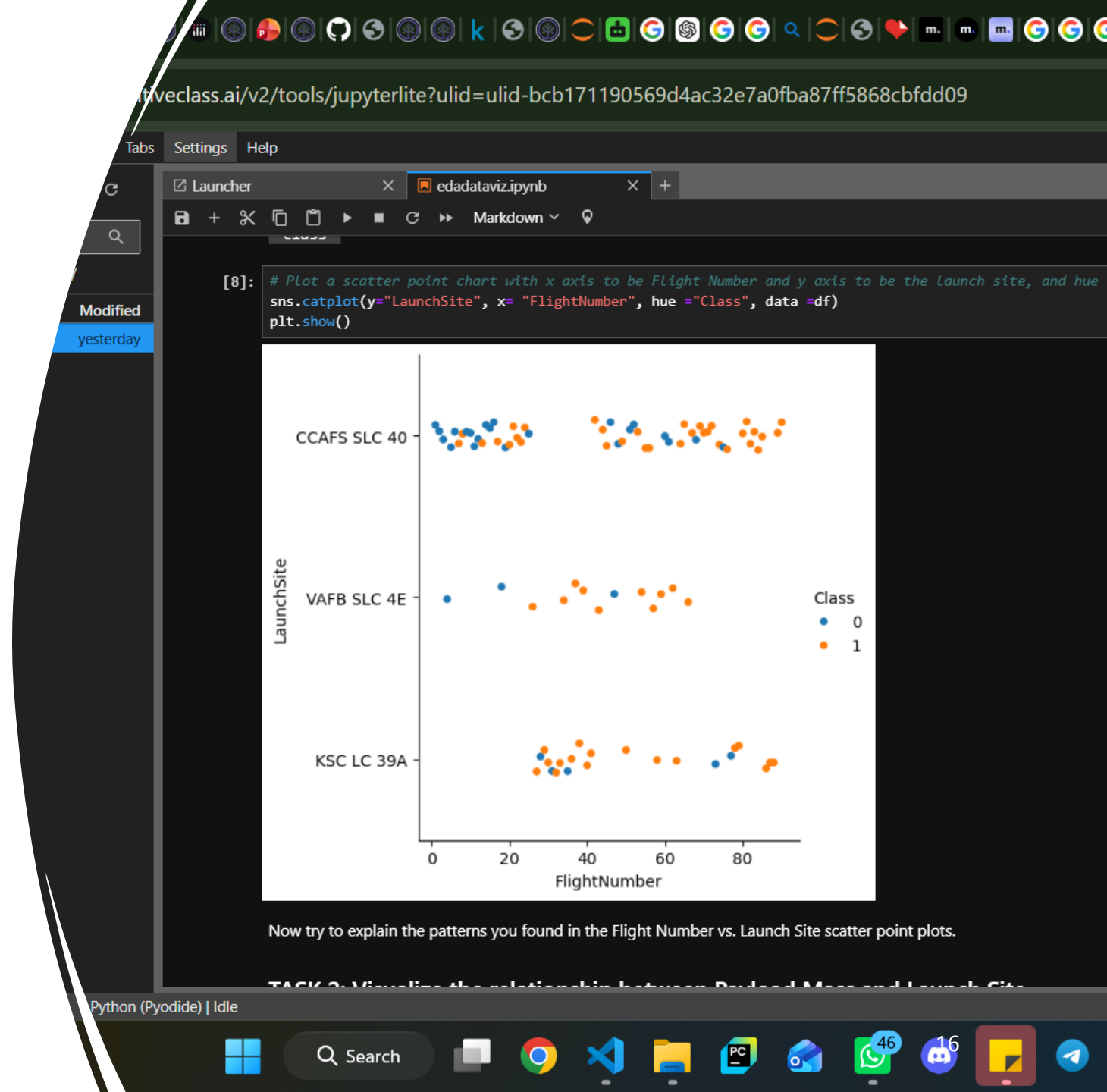
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

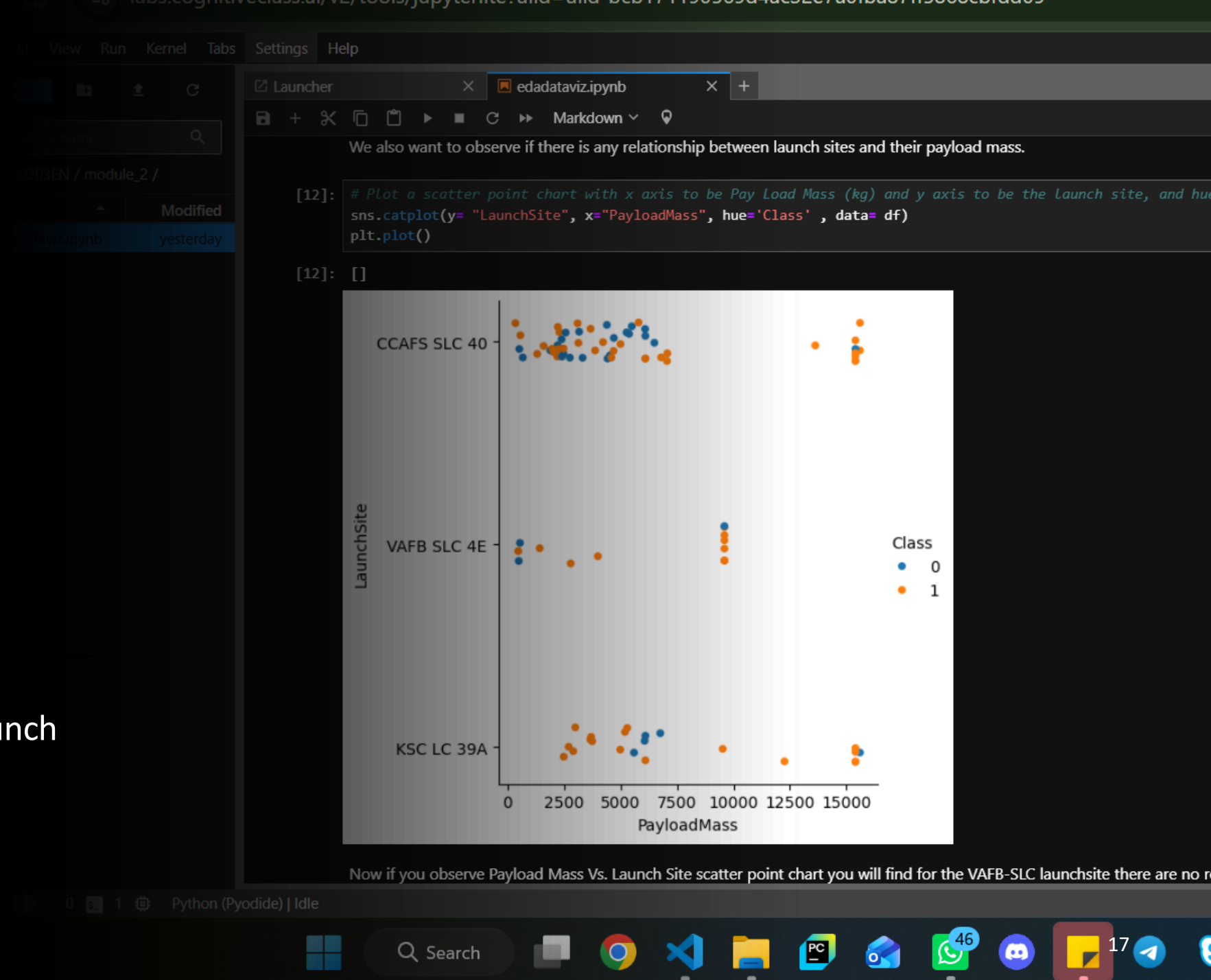
Flight Number vs. Launch Site

- a scatter plot of Flight Number vs. Launch Site



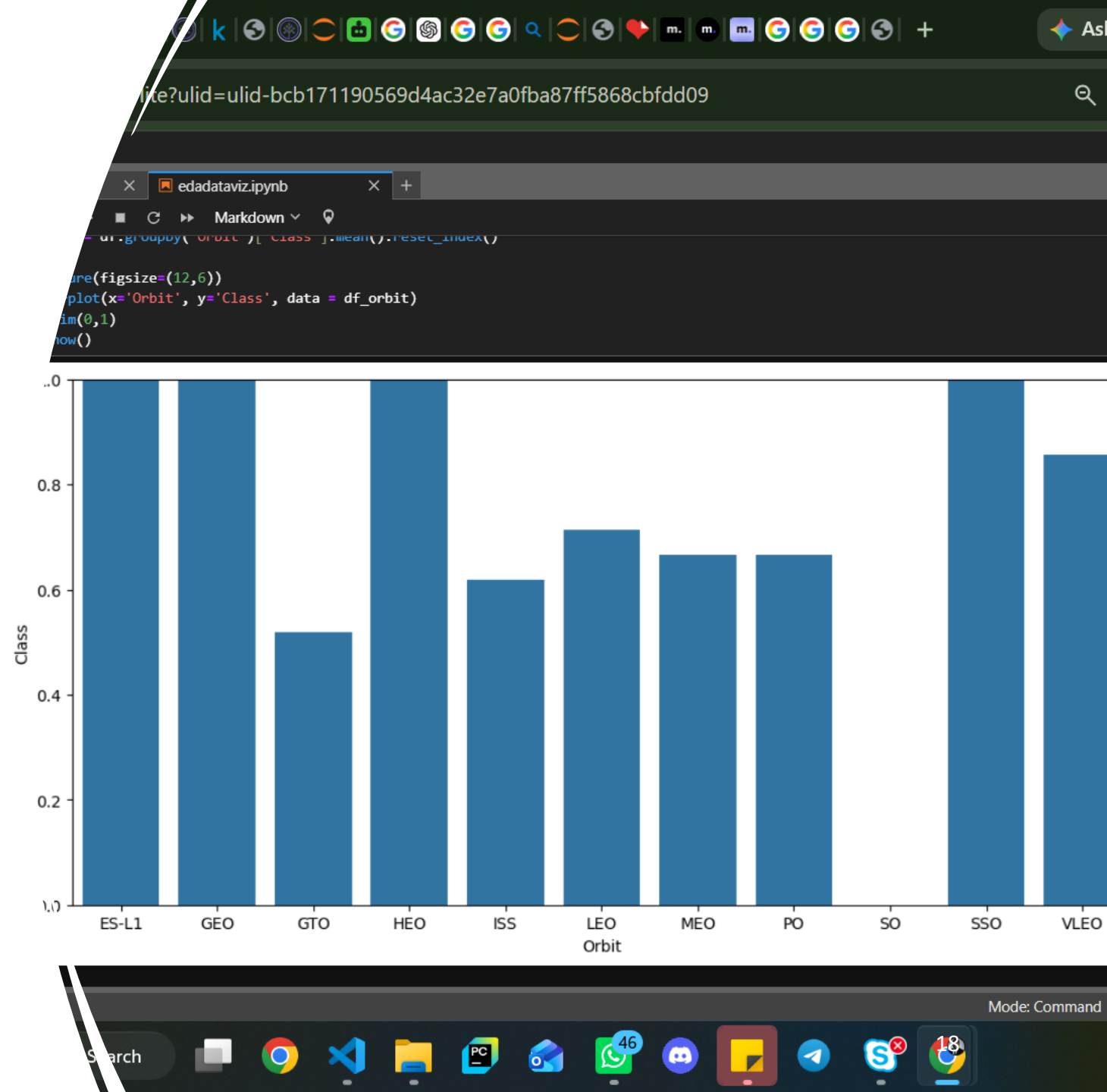
Payload vs. Launch Site

a scatter plot of Payload vs. Launch Site



Success Rate vs. Orbit Type

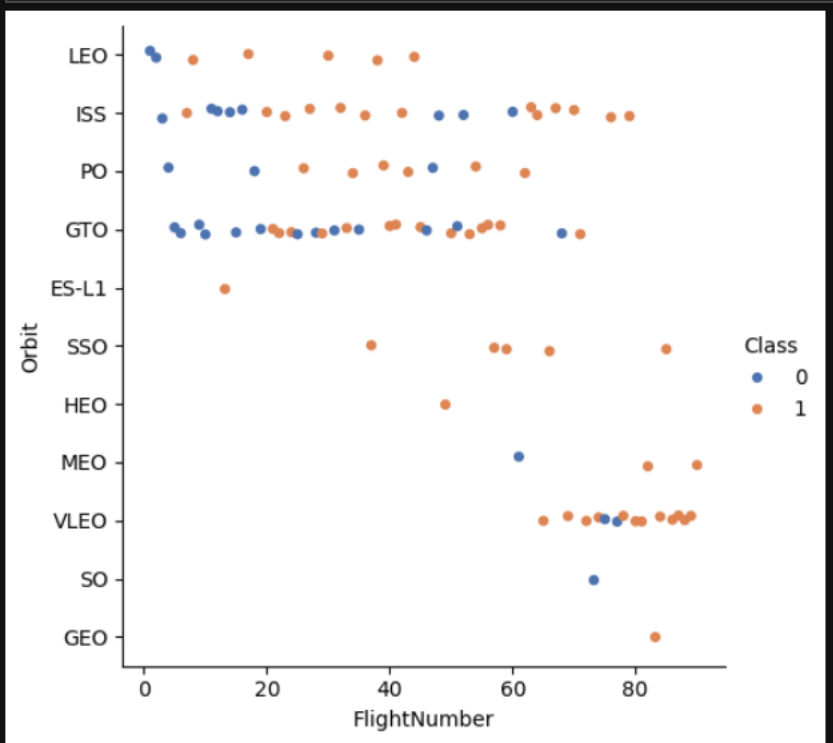
- a bar chart for the success rate of each orbit type



TASK 4: Visualize the relationship between FlightNumber and Orbit type

For each orbit, we want to see if there is any relationship between FlightNumber and Orbit type.

[21]: `# Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.catplot(x="FlightNumber", y="Orbit", hue='Class', data=df, palette='deep')
plt.show()`

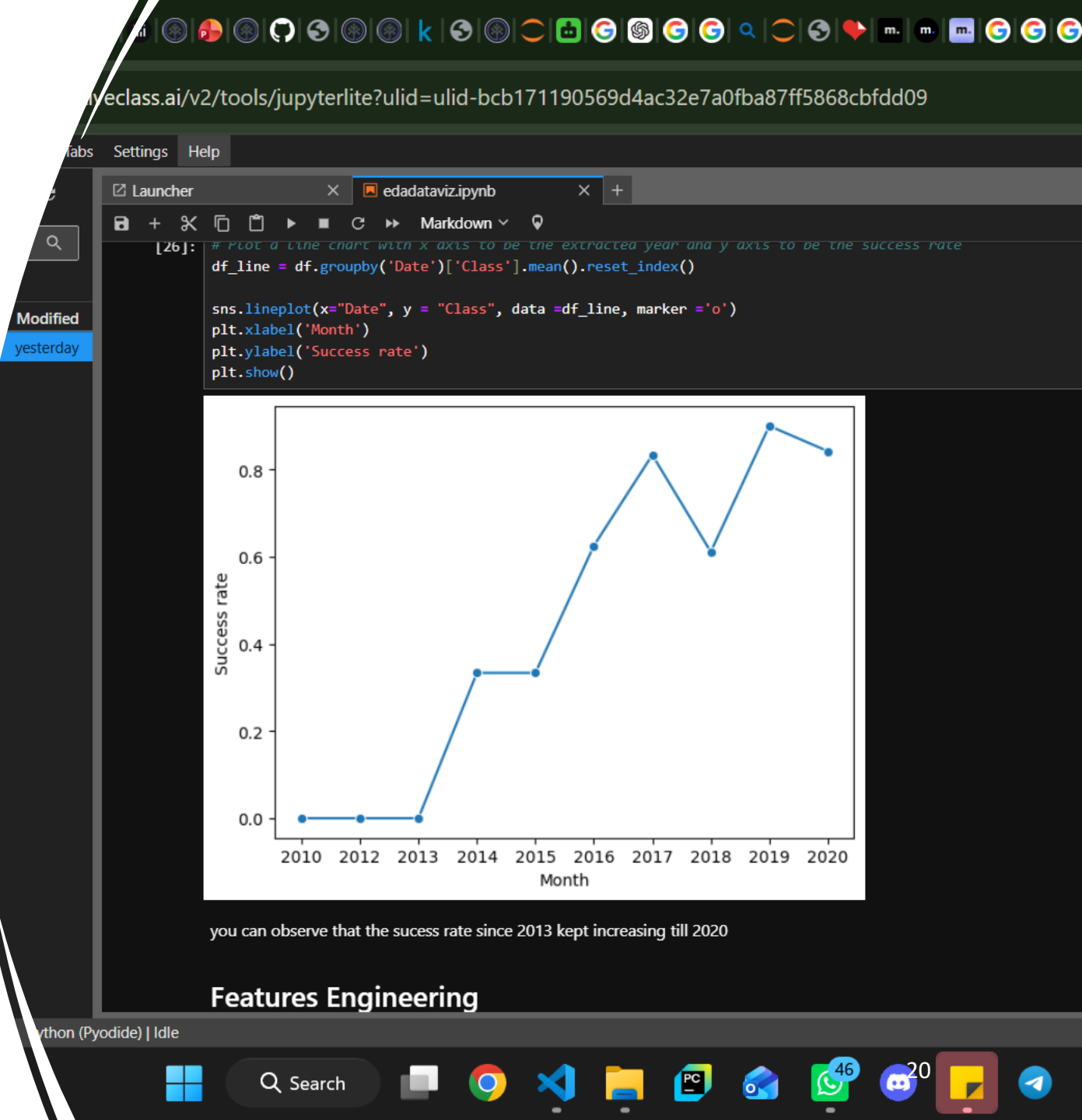


Flight Number vs. Orbit Type

- a scatter point of Flight number vs. Orbit type

Launch Success Yearly Trend

- a line chart of yearly average success rate



All Launch Site Names

- CAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

Total Payload Mass

- Query: `SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Customer" LIKE '%NASA (CRS)%';`

Result: total payload mass delivered for NASA Commercial Resupply Services missions 45,596 kg.

Average Payload Mass by F9 v1.1

- Query: `SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1'`

Result: average payload mass carried by the F9 v1.1 booster 2,928 kg.

First Successful Ground Landing Date

- Query: `SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';`

Result: 2015-12-22 — the date of SpaceX's first successful ground-pad landing (Orbcomm-2 mission).

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query: `SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000;`

Result: The query lists the specific F9 booster versions that successfully landed on a drone ship in this payload range

Total Number of Successful and Failure Mission Outcomes

- Query: `SELECT CASE WHEN "Landing_Outcome" LIKE 'Success%' THEN 'Success' ELSE 'Failure' END, COUNT(*) FROM SPACEXTABLE GROUP BY 1;`

Result: 61 successful landings vs. 40 failed/no-attempt outcomes across all recorded launches.

Boosters Carried Maximum Payload

- Query: `SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);`
- Result: lists the booster versions that carried the single heaviest payload recorded (~15,600 kg), mostly Starlink missions.

2015 Launch Records

- Query: filtered for year 2015 and "Landing_Outcome" = 'Failure (drone ship)';

Result: shows the 2015 drone-ship landing failures, their booster versions, and launch sites, this was early in the drone-ship recovery program, before the technique was perfected.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

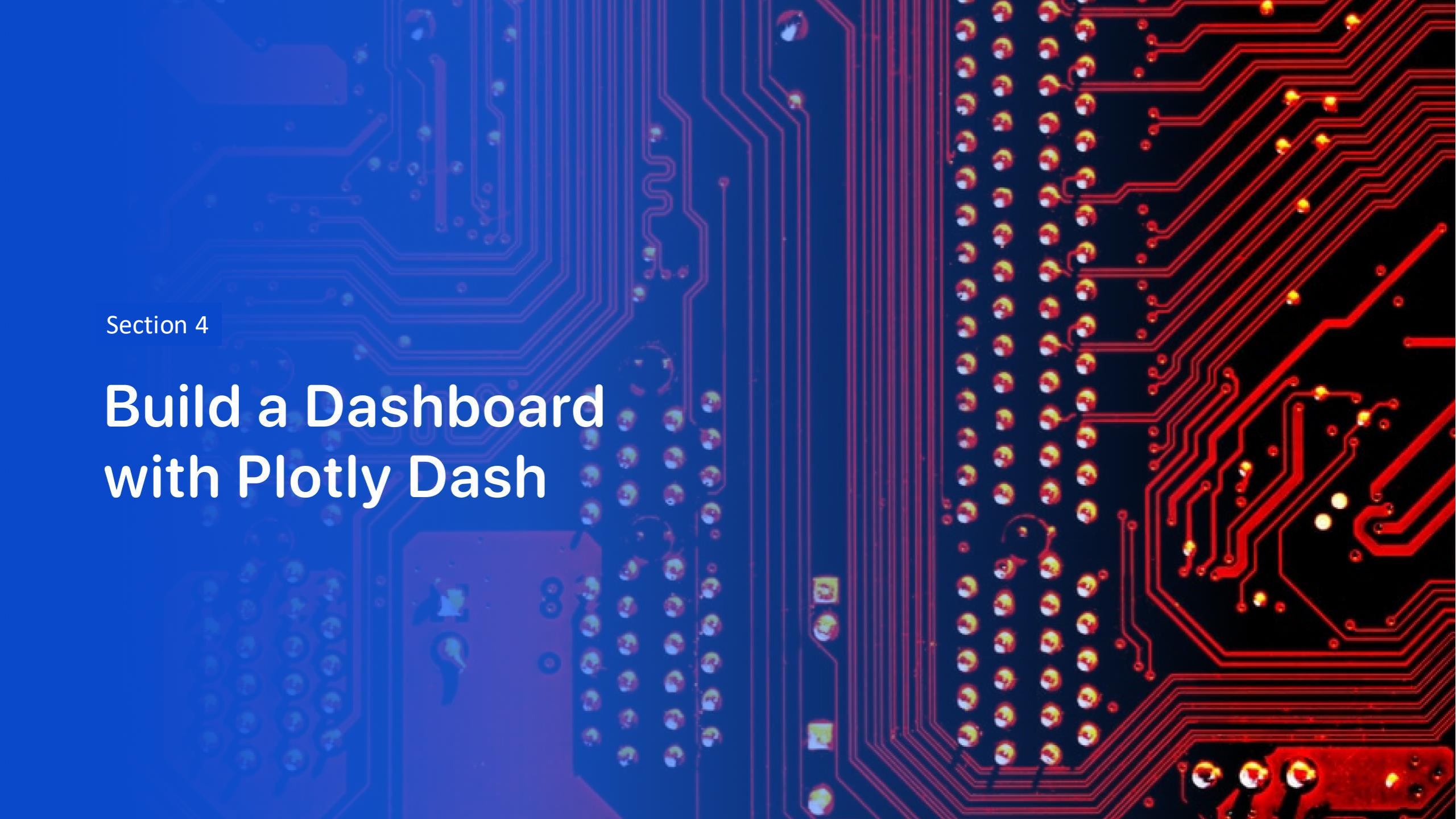
- Query: GROUP BY "Landing_Outcome" with COUNT(*), filtered between 2010-06-04 and 2017-03-20, ordered DESC;

Result: No attempt (10), Success/Failure drone ship (5 each), Success ground pad (3), Controlled ocean (3), Uncontrolled ocean (2), Failure parachute (2), Precluded drone ship (1).

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The horizon line of the Earth is visible, separating the dark surface from the blackness of space.

Section 3

Launch Sites Proximities Analysis



Section 4

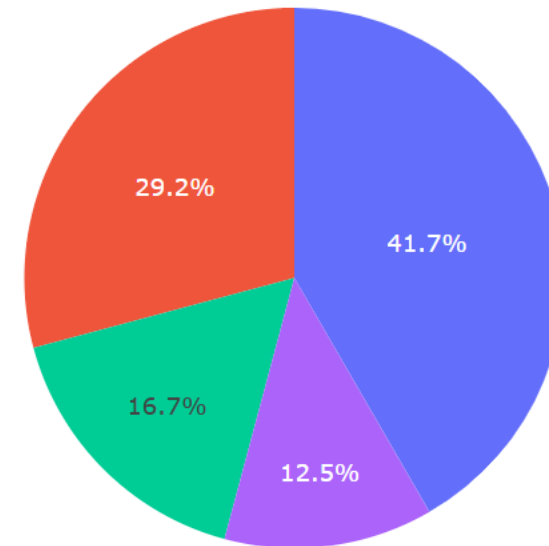
Build a Dashboard with Plotly Dash

Launch success count

- A screenshot of launch success count for all sites, in a piechart
- The chart shows the success rate launch sites

SpaceX Launch Records Dashboard

by Site

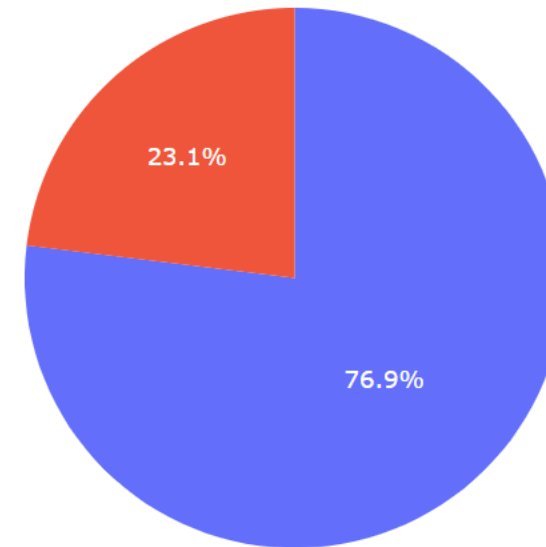


Highest success rate

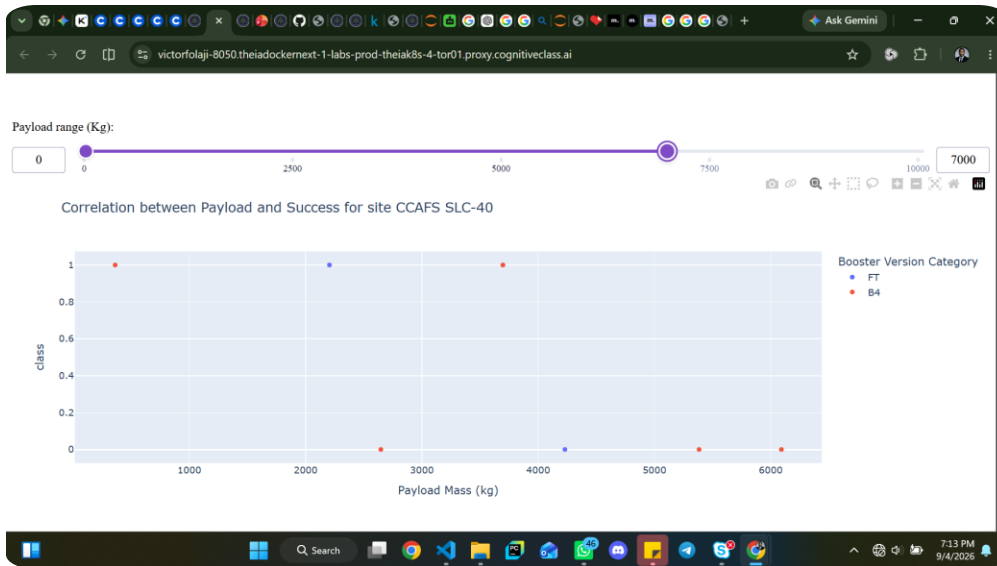
- A screenshot of the piechart for the launch site with highest launch success ratio
- KSC LC-39A is the site with the most success rate with 76.9% success rate

SpaceX Launch Records Dashboard

or site KSC LC-39A



Payload vs Launch scatter plot



- screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider

Section 5

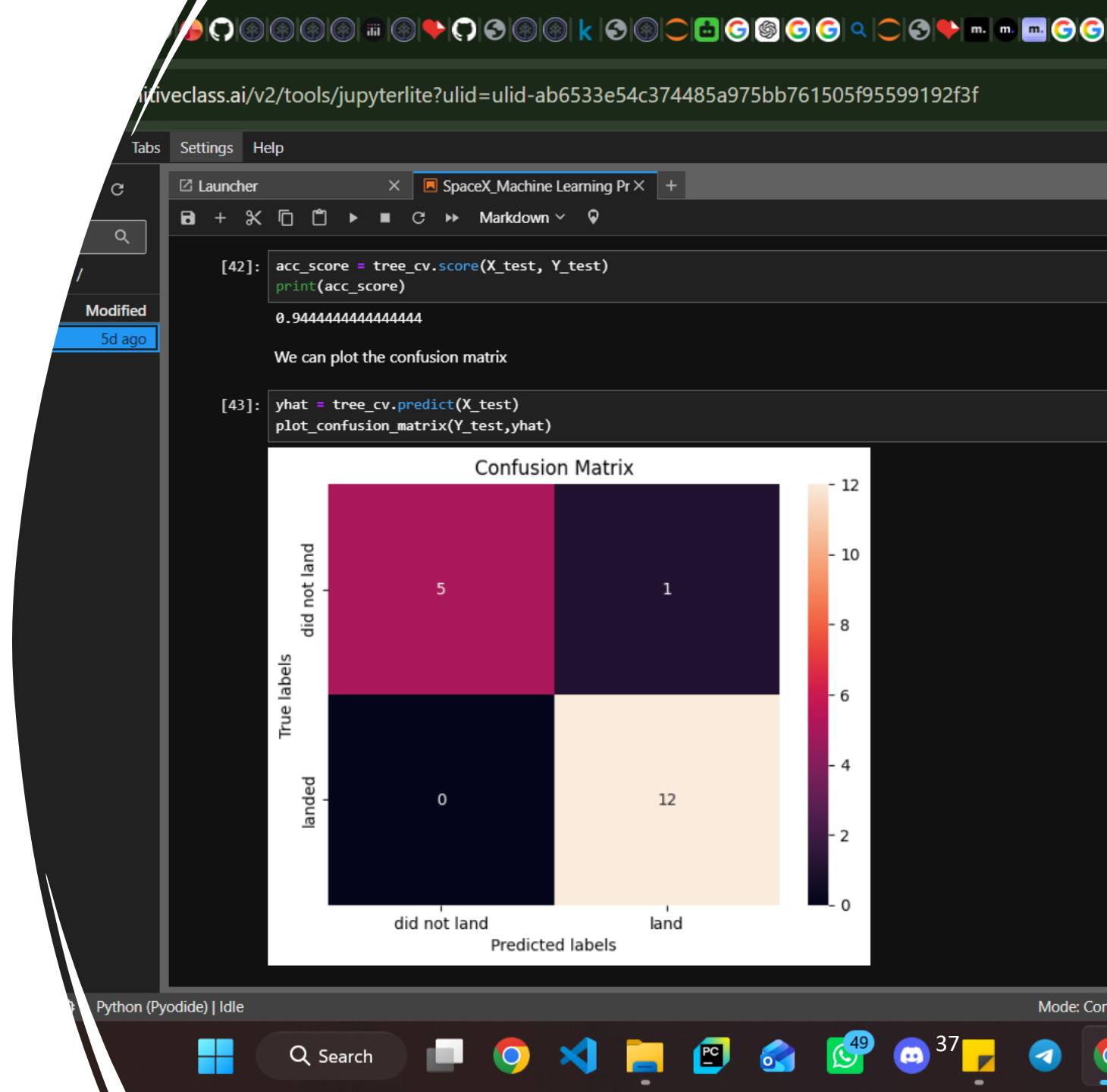
Predictive Analysis (Classification)

Classification Accuracy

- The Decision Tree classifier achieved the highest test accuracy (90.1%), $\approx$ narrowly ahead of Logistic Regression, SVM, and KNN (83.3% $\approx$ each).

Confusion Matrix

- The best model (Decision Tree) correctly classified 17 of 18 test launches: 12 true 'landed' predictions and 5 true 'did not land' predictions, with only 1 false



Conclusions

- PoiLanding success rate improved dramatically over time, from near 0% in 2013 to over 80% by 2020, driven by accumulated experience and technology upgrades.
- KSC LC-39A shows both a large number of successful launches and the highest success ratio among all four launch sites.
- Payload mass and orbit type jointly influence landing outcome which has heavier payloads succeed more at certain sites and orbits (e.g., LEO, VLEO) than others (e.g., GTO).
- The Decision Tree Classifier is the most reliable model tested, reaching ~88.9% test accuracy in predicting firststage landing success.
- Combining launch site, payload mass, and orbit gives more predictive power than any single variable alone — a useful insight for estimating launch costs when bidding against SpaceX.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

